

Statistics of Measurement (Lectures 5-9)

Lecture 5 - Hypothesis testing

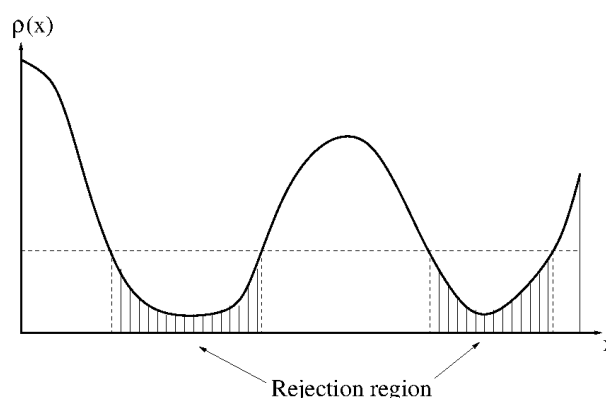
Heather Graven, 19 May 2023

1 Introduction

Now we turn to the estimation part of the course. The term ‘estimation’ means to deduce the values of parameters using data taken in an experiment. However, before we look at methods of estimation of the values of parameters, we will first look at a simpler problem. We will ask whether some data we have measured are consistent with a particular probability distribution. The probability distribution may be parameterless and hence fixed. However, it is more usual that there is a parameter with a particular value which we are interested in and we want to see if the data support it having this value. Hence, rather than trying to estimate the value of a parameter, we are seeing if this value is experimentally allowed. For example, we want to test if the parameter θ has the value θ_0 . This is our hypothesis and this is often labelled as H_0 . This is called ‘hypothesis testing’ as we are testing whether the hypothesis that the parameter has a particular value is consistent with data.

2 Hypothesis testing

The first problem we hit is what do we mean by ‘consistent’? Take the simplest case of having a hypothesised PDF $\rho_0(x)$ (which may depend on some parameter for which we have assumed a specific value) and making a single measurement, x_m . Basically, if the hypothesis is correct, we would expect the value we measure usually to be in the regions where the PDF is large. Hence, we would reject the hypothesis if the measured value is in regions where it is small. But how small is small? The absolute value of the PDF is not a useful measure; e.g. the value of the PDF in a uniform distribution is the inverse of the range so it can be arbitrarily small. We need an absolute probability so we must integrate the PDF over the regions where it is small. Quantitatively, a common way to do this is to choose a horizontal line, as shown in the figure, and integrate over ranges of the PDF which are smaller than the line. There can be one or several such ranges for a given horizontal line and together, all the values of x which lie within those ranges are said to be in the ‘rejection region’ (sometimes called the ‘critical region’).



The actual height of the horizontal line is not a particularly meaningful value, but it is usually chosen so that there is something like $\alpha = 10\%$, 5% or even 1% of the total probability

in the rejection region, where

$$\alpha = \int_{\text{Rejection region}} \rho_0(x) dx$$

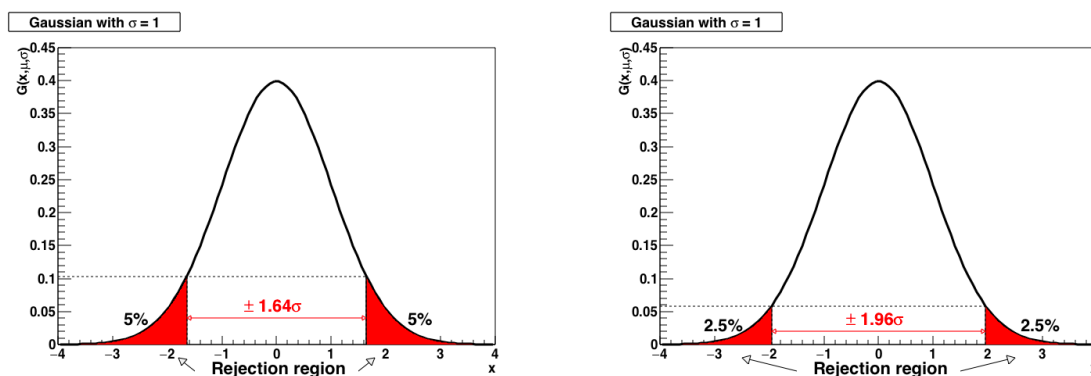
The fraction α is called the ‘significance level’ (for reasons we will discuss later) and is entirely up to us; there is no ‘right’ value. (As stated previously, estimation can be somewhat arbitrary.) This means whenever we quote the result of a hypothesis test, it is essential we give the significance level used.

Therefore, if we chose $\alpha = 0.05$ and measured x_m in the rejection region, we would say the hypothesis is false at the 5% significance level. Of course, we could have just been unlucky; by definition, 5% of the time we would expect to get x_m in the rejection region even if the hypothesis is true. Such an outcome, i.e. rejecting a true hypothesis, is called a ‘Type I error’ and is unavoidable for any finite significance level. The easy way to remember that this is one meaning of the significance level is that significance level equals ‘screw-up level’; we will screw-up 5% of the time if the hypothesis is true. Even at a significance level of 1%, then with 100 physicists testing the hypothesis of the theory of relativity, it is likely one would reject the theory. The other 99 might not bother publishing as their result is boring and expected, but you can be sure the one would (try to) publish their evidence that Einstein was wrong.

Therefore, we might want to reduce the significance level to avoid this. However, that makes the rejection region smaller and so it is less likely we will be able to reject the hypothesis if it is false. Hence, we have to choose values which set a balance between these effects.

Note, we can never prove a hypothesis is true. Even if x_m was right at the peak of the PDF, it does not mean that the PDF we have chosen is correct. Hence, all we can do is reject what we consider to be false hypotheses. If we get an x value which will not reject H_0 , even though it is actually false, it is called ‘Type II error’. Unless we actually know the true PDF (given that the hypothesis under test is false in this case), we cannot evaluate the probability of a Type II error.

Because of the CLT, the Gaussian distribution is the most common, and the integrals over various ranges were given in Lecture 3. Some plots are shown below for a total rejection region of 10% and 5% (which have 5% and 2.5% in the tail on each side). Hence, if you had a theory which predicted a Gaussian distribution centred on zero for some quantity, you would reject the theory at 5% significance level if the value you measured was more than 1.96σ away from zero.



In fact, you will often hear the significance level quoted in terms of number of sigma rather than percentages, as a Gaussian PDF is so common. Indeed, this explains why it is called ‘significance level’. Clearly, if we pick a 5% significance level, which is outside 1.96σ , and then find x_m is in the rejection region, this is a more meaningful result than if we had picked a 10% significance level at 1.64σ and found x_m in the rejection region in that case. Hence, a rejection with significance level 1.96σ is more important, i.e. statistically more significant, than one at

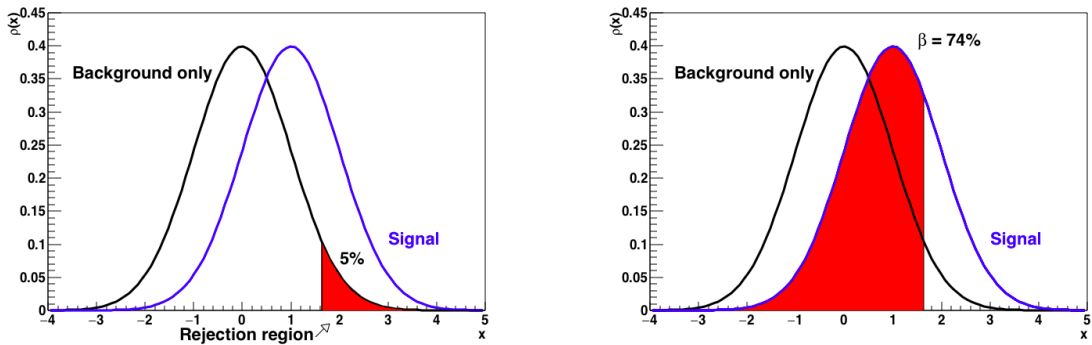
1.64σ . Note, this is even done for non-Gaussian PDFs, where a significance level is converted into an equivalent of sigmas using the Gaussian integrals, even though the actual PDF has no parameter called sigma.

3 The alternative hypothesis

So far, so straightforward. However, even in the above, which seems intuitive, there are some hidden assumptions which we need to understand. The basic issue is that, if we have a hypothesis which we are testing to see if it is false, then we have to think about what could be the true hypothesis. There must always be *some* PDF which is true, if it is not the one we are testing. Hence, it is necessary to have an ‘alternative hypothesis’ which we compare against. The choice of the alternative can change the rejection regions for our hypothesis under test completely.

A very simple case; if we measure x_m to be in a range which has a low PDF for our hypothesis, but it has an even lower value for the alternative, then is it still correct to reject the hypothesis? H_0 is unlikely, but more likely than the other possibility. Hence, for such a case, we would probably not want the rejection region to contain those values of x .

A specific example; we want to see if there is a signal of some physical process predicted by a new theory. If there is no signal, then x will have a Gaussian distribution centred at 0. However, if the new theory is true, then x will have a Gaussian distribution of the same width centred at 1. The former is often called the ‘null hypothesis’ (hence H_0) as it means nothing new is happening, and the other is the alternative hypothesis H_1 with PDF $\rho_1(x)$. The PDFs for these are shown below. Note, the null hypothesis is the one we are testing, as we cannot prove the new theory is correct; there could be many other theories which give something similar. All we can do is try to reject the null hypothesis and so say there is a need for a some new theory. Clearly, if we measure a large positive value of x_m , then we can reject H_0 . However, if we measure a large negative value, then even if it is many σ away from zero, there is still a higher probability for H_0 than H_1 . Hence, we must pick our rejection region only to include positive values of x . Therefore, the 5% rejection region is now everything above 1.64σ ; note *not* 1.96σ as we have a one-sided integral, not a two-sided one in this case. Hence, this alternative hypothesis has changed our rejection region significantly.



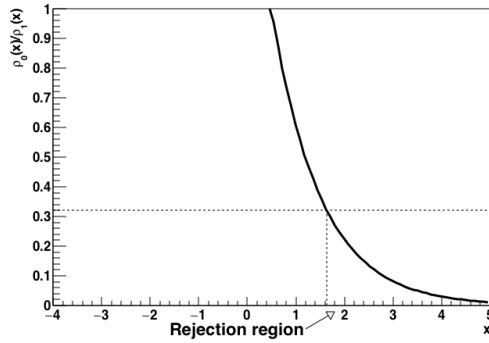
The other thing a well-defined alternative hypothesis allows us to do is find out how often we will not reject H_0 , even if it is false, i.e. how often we will have a Type II error. If H_0 is false, then by definition, we assume H_1 is true. Therefore, we can integrate the PDF for H_1 over all values of x which are *not* in the rejection region. This tells us the probability β of getting an x value which will not reject H_0 , even though it is false, i.e. the probability of a Type II error.

$$\beta = \int_{\text{Not rejection region}} \rho_1(x) dx$$

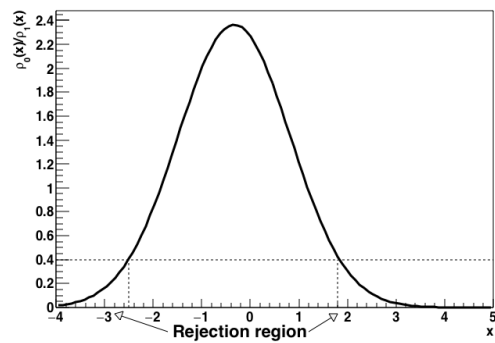
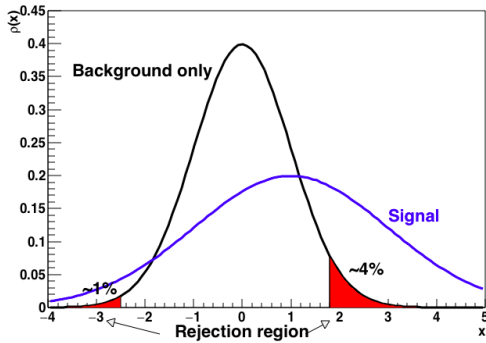
The probability $1 - \beta$ is called the ‘power’ of the test. For the above example

$$\beta = \int_{-\infty}^{1.64} G(x; 1.0, 1.0) dx = 0.74$$

Ideally, we would like to minimise both α and β . It turns out for cases where both the hypothesis under test and the alternative hypothesis are well-defined, as above, then the best rejection region selection is not $\rho_0(x) < \text{some value}$, but is $\rho_0(x)/\rho_1(x) < \text{some value}$, i.e. the ratio of the two PDFs. By ‘best’ it means the selection which minimises α and β as far as possible. This can be proved in general for any two PDFs and is called a ‘Neyman-Pearson test’. Hence, our intuitive use of which PDF was higher is indeed a sensible choice.



A final comment; most text books simply state that the alternative hypothesis defines whether you do a one-sided or two-sided integral. These are indeed the most common cases. However, there can be others; for example if the signal Gaussian PDF above had a bigger width (due to extra noise predicted as occurring in the new signal process) then at some large negative value of x , H_1 would become more probable than H_0 again. Hence, the rejection region would be two-sided, but would be a very uneven, with most at large positive x and a smaller amount at large negative x .



4 Composite alternative hypotheses

It is often the case that the alternative hypothesis is not a well defined PDF. For the example above, the new theory could predict a signal will be higher than zero, but the actual size of the signal is not known. Hence, the Gaussian for H_1 could have any mean as long as $\mu > 0$. We

cannot therefore write down an explicit PDF for H_1 and so we cannot find the power β , nor do an optimal Neyman-Pearson test. This is called a composite hypothesis.

In this case, the signal is predicted only to be higher than zero, not lower, so we know the PDF for H_0 is higher than for H_1 when $x \leq 0$, so that we would again use a one-sided test. This is often the case in physics. However, for some cases, the alternative hypothesis can be equally likely to shift the value up or down, and then we need the two-sided rejection regions shown initially. Hence, for the first part of the lecture we had been implicitly assuming a rather ill-defined and very general alternative hypothesis; in these cases, we can take a uniform distribution as the alternative hypothesis to define our rejection region.

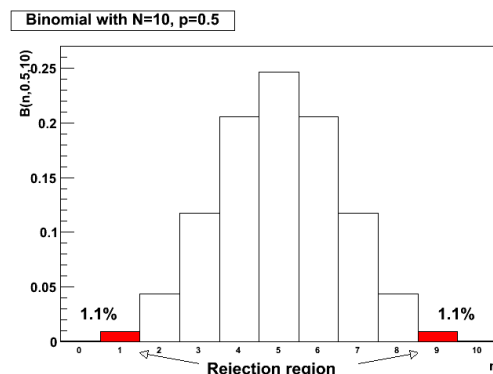
5 Discrete distributions

All of the above maps over from continuous PDFs to discrete distributions except for one complication. Due to the non-continuous nature of the probabilities, then it is usually not possible to find a region which contains exactly our chosen significance level.

For example, I flip a coin 10 times; is the result consistent with the coin being unbiased? The distribution is binomial and unbiased means the probability of heads p is equal to the probability of tails, so $p = 0.5$. Hence, I have chosen a particular value of the binomial parameter for my null hypothesis H_0 . My alternative hypothesis is that the coin is biased, either towards heads or tails, so $p > 0.5$ or $p < 0.5$. Hence, I want a two-sided rejection region and I choose a significance level of 5%. Assuming H_0 is true, the probability of getting ten heads or ten tails is $p^{10} = 1/1024$, of nine heads or nine tails is $10p^9(1-p) = 10/1024$, and of eight heads or eight tails is $45p^8(1-p)^2 = 45/1024$. Hence, a rejection region of ten heads, ten tails, nine heads and nine tails totals $22/1024 = 2.1\%$, which is below 5%, while also including eight heads and eight tails would make it $112/1024 = 10.9\%$, which is above 5%.

How do we choose the rejection region in this case? Consider the continuous case again. Say we choose a 5% rejection region and find x_m falls within it. If we had instead chosen a 10% rejection region, this would completely include our previous 5% rejection region and so the measured x_m would also reject the hypothesis at 10%. Hence, a rejection at a lower significance level can also be interpreted as a rejection at a higher significance level. Another way to see this is that for the 5% rejection region, the actual x_m could be within e.g. the range which would be within a 1% rejection region. Hence, it is conservative but correct to label a lower significance level with a larger value.

Hence, for our coin example, we would choose the smaller rejection region, i.e. nine and ten heads and tails, for our 5% significance level, even though it contains only 2.1% of the probability.



6 Setting significance from the data

The handling of significance levels for discrete distributions discussed above is clearly conservative and so in reality, you might think it would be more sensible to choose our significance level to correspond to one of the possible values, e.g. either 2.1% or 10.9%, so that we are not being too cautious. Indeed, this is very common in practise; rather than setting the rejection region before doing the experiment, we do the measurement and then find out the smallest significance level which would include x_m in the rejection region. We then say we can reject the hypothesis at that significance. The significance level determined by the data is called the p value, and so we set α equal to the p value. (Obviously, if it is large, then the result does not have much to say about the hypothesis.) For example, if we got ten heads from ten coin tosses, then we would have a p value of 0.002 and we could reject the unbiased coin hypothesis at 0.2% significance level. Using the convention of quoting the number of Gaussian standard deviations corresponding to the integrated probability, this corresponds to 3.1σ . As we can see, significance levels of 3σ or more are very unlikely and hence rejections based on these are considered very significant.

Statistics of Measurement

Lecture 6 - Point estimation and maximum likelihood

Heather Graven, 22 May 2023

7 Introduction

A common issue in physics is that we have a distribution for which the parameters are not known and we do an experiment to try to determine them. This deduction of the values of the underlying parameters is called “estimation” and is the main aim of this course; it is what is often meant by the study of statistics. In the last lecture, we saw that a hypothesis test often asks the question “Is a particular value of the parameter consistent with the data?” We will now ask the question “What value of the parameter is most consistent with the data?”, i.e. we want the ‘best estimate’ of the parameter. We saw in the last lecture that we would expect measurements to be in the regions where the PDF is largest. This leads us to the concept of maximum likelihood. Note, “likelihood” here is used in a technical sense; as we will see, it is different from probability, even though the two are used interchangeably in non-technical everyday speech. This is similar to “energy”; it means something a lot more specific in physics than the way people use it normally.

8 Estimation

Finding the ‘best’ estimate is called ‘point estimation’ as in general, for many parameters, we are determining a best point in the multidimensional parameter space. Clearly, we will also be interested in the uncertainty on the best estimate and that is discussed in Lecture 8.

There are many ways to estimate a value for a parameter from a set of experimental measurements. At a basic level, any point estimation method boils down to finding a function M of the measurements x_i which gives an estimate of the value of the parameter θ

$$\hat{\theta} = M(x_1, x_2, \dots, x_N)$$

We write the estimate of the parameter as $\hat{\theta}$ to distinguish it from the actual (but unknown) value θ . Note, $\hat{\theta}$ is also a random variable and hence has a PDF. All estimation comes down to determining what method to use, which defines what the function M is.

You will have already met some methods for estimation; one is so obvious that it is easy to forget that it is formally an estimate at all and that is the mean of a set of measurements x_i

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

so here the function M is simply the sum divided by N .

There are many methods of estimation; some are good and some are not. Good estimates are ones which are

1. **Consistent:** meaning, in the limit of large N , that $\hat{\theta} \rightarrow$ the true value of θ .
2. **Unbiased:** meaning that $E(\hat{\theta}) =$ the true value of θ .
3. **Efficient:** meaning that $V(\hat{\theta})$ is small.

There is often a trade-off between these characteristics, e.g. you can use a method which is biased but efficient, or unbiased but has a larger variance and so is less efficient.

The above estimate $\hat{\mu}$ of the average is very widely used and that is because it is consistent, unbiased, and has good efficiency. However, it is of course limited to finding the mean, not general parameters of a distribution. In this course, we will look at two other main methods of parameter estimation; the method of maximum likelihood and the chi-squared (χ^2) method. These are more general methods which can be used to find arbitrary parameters of a distribution. In this lecture, we will look at the maximum likelihood method, which can be applied to any probability distribution. In the next lecture, we will study the chi-squared method, which is applicable when measurements are made of variables with (at least approximately) Gaussian distributions; this is of course a very common case due to the CLT. While less general than the maximum likelihood, it is often simpler to implement. In both cases, these methods are usually consistent and have good efficiency, but are only unbiased for large N .

9 The likelihood function

It is very common that we believe we know the form of the probability (density) distribution $P(x; \theta_i)$ for an experiment but this depends on unknown parameters θ_i . The point of doing the experiment is then to determine these parameters. Here we will consider the one parameter case for clarity but the method is easily generalised to more than one parameter.

Imagine we have taken N independent data points with our experiment so our data values x_i are now fixed. Because they are uncorrelated, the total probability is the product of the separate probabilities, so we can write

$$L(\theta; x_i) = \prod_{i=1}^N P(x_i; \theta)$$

which defines the “likelihood” function L for our experiment. Since the data values x_i are fixed, L is purely a function of the parameters θ_j and indeed, it is often just written as $L(\theta)$ to emphasise this. The likelihood is effectively the probability of seeing the data values actually observed, as a function of the parameters. Note, it is *not* a PDF for θ .

For example; say we were performing 10 binary outcome trials and saw 3 of one of the two outcomes. The probability distribution for this experiment is the binomial and we want to estimate the probability p for a single trial (the equivalent to θ above). The likelihood for this experiment has only one term in the product (as we only did one set of trials) and so is

$$L(p; 3, 10) = B(3; p, 10) = \frac{10!}{3!7!} p^3 (1-p)^7 = 120 p^3 (1-p)^7$$

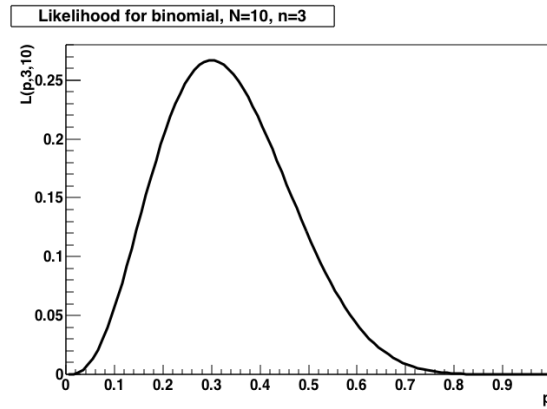
where it is clear it is a function of p only. This function is plotted below.

The obvious estimate for p would probably be the average fraction of the outcome which was observed, i.e. $\hat{p} = 3/10 = 0.3$. As is seen in the figure, the likelihood does indeed peak at that value. This implies that the value of p which gives a maximum in the likelihood function, i.e. which would make the probability of seeing the observed data the largest, is a sensible estimate.

In practice, for reasons which will become clear later, we always take logarithms of both sides of the original equation, giving the “log-likelihood”. This turns the product into a sum

$$\ln[L(\theta)] = \ln \left[\prod_{i=1}^N P(x_i; \theta) \right] = \sum_{i=1}^N \ln[P(x_i; \theta)]$$

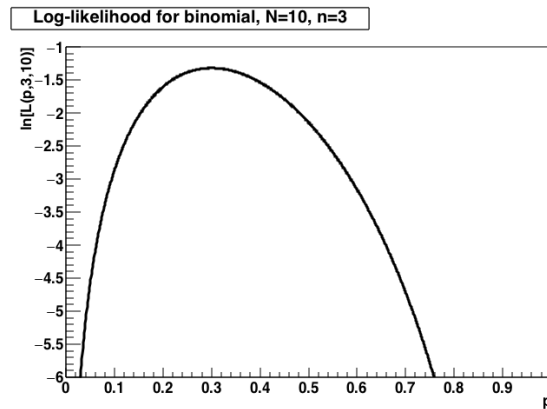
which is usually easier to handle mathematically. Since the logarithm is a monotonically increasing function, the maximum of L is also the maximum of $\ln(L)$ and so we can work with



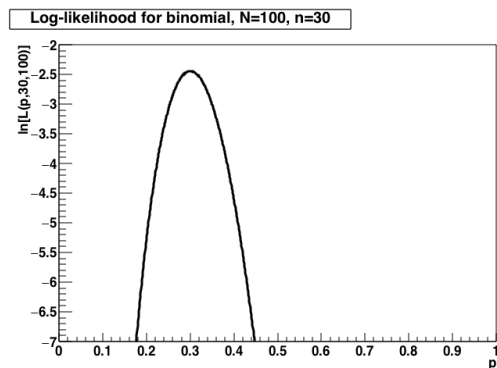
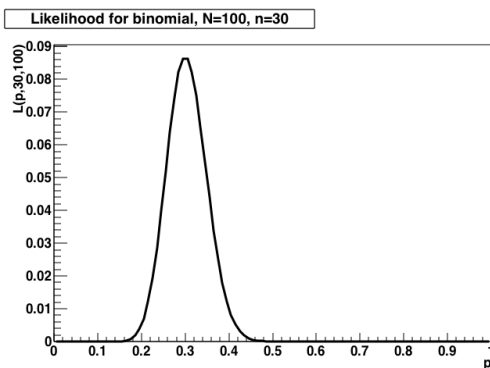
$\ln(L)$ for all calculations. The log-likelihood for the binomial above is

$$\ln[L(p; 3, 10)] = 3 \ln(p) + 7 \ln(1 - p) + \ln(120)$$

This is shown below and it does indeed peak at the same place.



Furthermore, if we had seen 30 outcomes out of 100 trials (rather than 3 out of 10), then we would expect our estimate $\hat{p}$ to still be 0.3 but also to be more accurate. The likelihood and log-likelihood curves for this case are shown below. Again they peak at $p = 0.3$ but also it is clear they are more sharply peaked around this value. Hence, we would expect the width of the likelihood to reflect the error on the estimate. We will come back to this in Lecture 8.



10 The maximum likelihood principle

This idea can be elevated to the level of a principle; the values of the parameters which maximise the log-likelihood function are our best estimates of those parameters. Basically, we want the measurement value to be at the peak of the PDF so we adjust the PDF parameters to make this happen. By a good estimate here, we mean it turns out they are consistent and efficient, and unbiased at least in the limit of large numbers of data points. Hence, this gives us a general method for estimating parameters of a probability distribution.

Clearly we do not need to make a plot each time we have a log-likelihood function; to find the maximum we take a derivative and find the values of the parameters which make it zero. Any method of adjusting the parameters to maximise (or minimise) a value and hence make an estimation is generally called ‘fitting’ for the parameters. E.g for the binomial above

$$\frac{d \ln(L)}{dp} = \frac{3}{p} - \frac{7}{1-p}$$

The estimate $\hat{p}$ is given by this being zero, for which

$$\frac{3}{\hat{p}} = \frac{7}{1-\hat{p}} \quad \text{so} \quad 3 - 3\hat{p} = 7\hat{p} \quad \text{so} \quad \hat{p} = \frac{3}{10}$$

as we would expect.

11 Gaussian approximation

Let’s look at the Gaussian case in some detail. The probability distribution for a single measurement x is

$$G(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{so} \quad \ln[G(x; \mu, \sigma)] = -\ln(\sigma\sqrt{2\pi}) - \frac{(x-\mu)^2}{2\sigma^2}$$

If σ is known, then one measurement x_1 allows us to estimate μ . The log-likelihood is

$$\ln[L(\mu)] = -\ln(\sigma\sqrt{2\pi}) - \frac{(x_1 - \mu)^2}{2\sigma^2}$$

which obviously peaks at $\hat{\mu} = x_1$ as might be expected. The log-likelihood function is a quadratic in μ around x_1 .

However, a Taylor expansion of any function around a maximum (where the first derivative is zero) will approximate to a constant plus a negative quadratic term, just as above. Hence, any PDF will look approximately Gaussian close to the peak. This ‘Gaussian approximation’ will be used in Lecture 8. An example of this for the binomial case discussed previously is shown in the plots below.

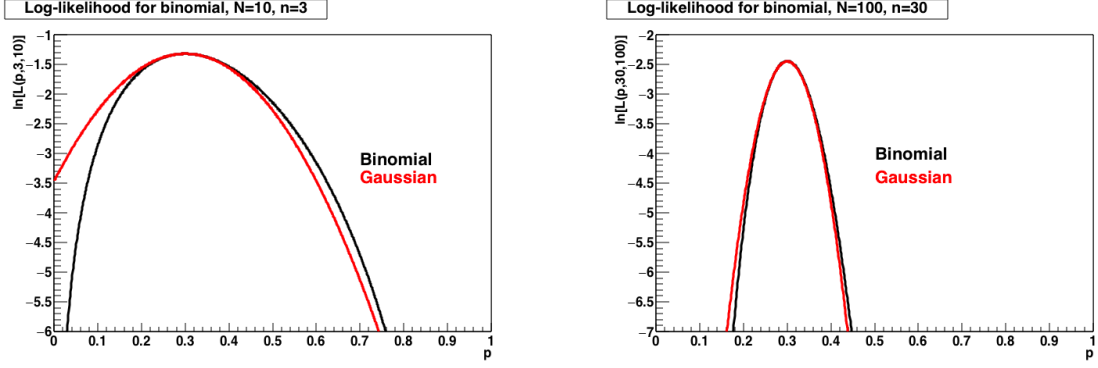
12 Gaussian parameter estimation

Now take the case of N measurements of x_i which has a Gaussian probability distribution with unknown mean μ and width σ . The probability distribution for a single measurement x_1 is

$$G(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{so} \quad \ln[G(x; \mu, \sigma)] = -\ln(\sigma) - \frac{1}{2} \ln(2\pi) - \frac{(x-\mu)^2}{2\sigma^2}$$

Hence, the total log-likelihood function is

$$\ln[L(\mu, \sigma)] = \sum_{i=1}^N \ln[G(x_i; \mu, \sigma)] = -N \ln(\sigma) - \frac{N}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2$$



We need to maximise with respect to both μ and σ in this case, so we need both derivatives

$$\frac{\partial \ln(L)}{\partial \mu} = \frac{1}{2\sigma^2} \sum_{i=1}^N 2(x_i - \mu), \quad \frac{\partial \ln(L)}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^N (x_i - \mu)^2$$

to be zero. We can use the first of these being zero to find the estimate on μ

$$\sum_{i=1}^N (x_i - \hat{\mu}) = 0 \quad \text{so} \quad \sum_{i=1}^N x_i = N\hat{\mu} \quad \text{i.e.} \quad \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

as we stated earlier. Solving for the estimate of σ using the other derivative

$$\sum_{i=1}^N (x_i - \hat{\mu})^2 = N\hat{\sigma}^2 \quad \text{so} \quad \hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$$

You may know this is a biased estimate; it turns out the expectation value $E(\hat{\sigma})$ is less than the true value of σ . The unbiased estimate is

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu})^2$$

which differs by the ‘Bessel correction’ factor of $N/(N-1)$, which clearly goes to 1 for large N . Hence, this is an example of a case where the maximum likelihood gives an unbiased estimator only in the large N limit.

When using maximum likelihood estimation, one must be careful of biased estimates, particularly at small N . This can be explored by calculating the expectation value of your parameter and checking for bias.

Statistics of Measurement

Lecture 7 - The chi-squared estimation method

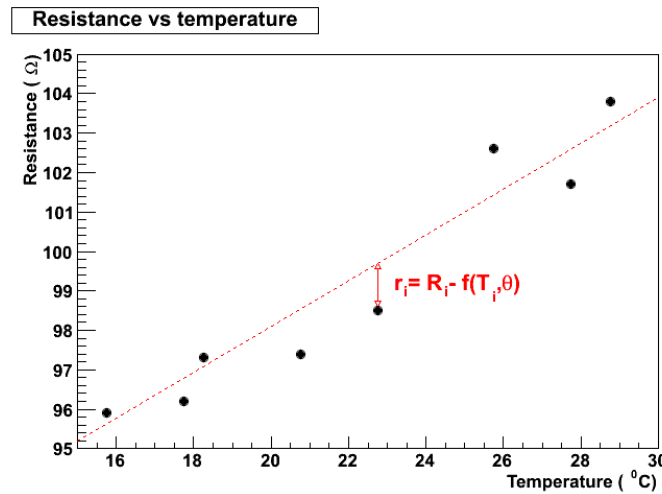
Heather Graven, 23 May 2023

13 Introduction

We saw the maximum likelihood method in the previous lecture. Today, we will look at the chi-squared method, which is applicable in the case of measurements made from an underlying Gaussian distribution. The chi-squared is closely related to the least squares method, so we will start with that.

14 Least squares

The chi-squared method is a more powerful generalisation of the method of least squares. The most common application of least squares is when we have a set of measured data values y_i which were taken while another variable x_i was varied. The idea is that the true values of y depend on x according to some functional form (or ‘model’) $y = f(x; \theta_1, \theta_2, \dots) = f(x; \theta_j)$ and that we want to find the parameters θ_j of this function, where j runs over the number of parameters. One example would be to estimate the linear temperature dependence of a resistor. For this, the function we would use is a straight line $f = \theta_1 + x\theta_2$ and so there are two parameters. Measurements of the resistance are taken for various temperatures and the results of the resistance as a function of temperature are plotted below.



To do this estimation, we use the “residual” r which is the difference of each measured value from the function value

$$r_i = y_i - f(x_i; \theta_1, \theta_2, \dots) = y_i - f(x_i; \theta_j)$$

We then take the sum of the squares of the residuals

$$S(\theta_j; y_i) = \sum_{i=1}^N r_i^2 = \sum_{i=1}^N [y_i - f(x_i; \theta_j)]^2$$

The value of S is considered as a function of the parameters θ_j given that we have observed data values y_i . Note, the x_i are taken as known values with no (or negligible) uncertainties. We

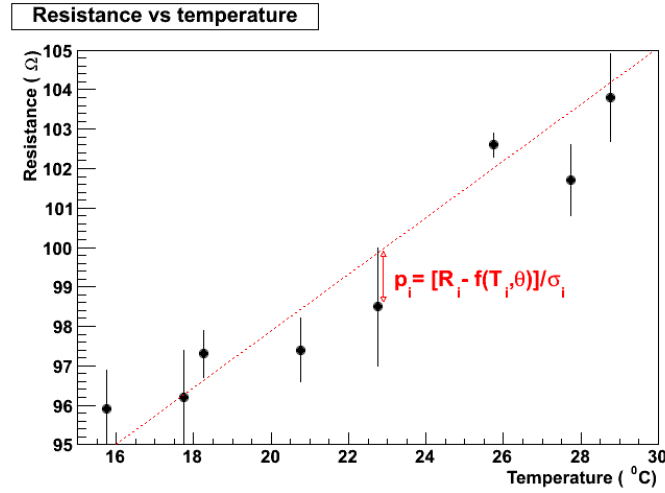
then adjust the parameters so as to minimise this sum; mathematically we need to solve the simultaneous equations

$$\frac{\partial S}{\partial \theta_j} = 0$$

where there is one equation for each parameter θ_j . This gives us the estimates $\hat{\theta}_j$. Clearly this minimisation will tend to make the function agree with the measured values as far as possible, so that the differences, i.e. the residuals, are small. Hence, it is intuitive that this will give a reasonable method for estimating the θ_j .

15 Chi-squared method

The chi-squared method is a more rigorous approach than least squares but is fundamentally similar. The difference is that it incorporates the uncertainties on the measurements y_i , here denoted by σ_i , which can be different for each point. Clearly, measurements with large uncertainties should have less power than those which are more accurate and have small uncertainties. Note, this of course requires that the uncertainties are already known.



The chi-squared method starts by dividing the residual by the uncertainty to form a quantity called the “pull” p_i

$$p_i = \frac{r_i}{\sigma_i} = \frac{y_i - f(x_i; \theta_j)}{\sigma_i}$$

If all the data points are independent, the sum of the squares of the pulls is then called the chi-squared

$$\chi^2(\theta_j; y_i) = \sum_{i=1}^N p_i^2 = \sum_{i=1}^N \left[\frac{y_i - f(x_i; \theta_j)}{\sigma_i} \right]^2$$

and we again want the minimum, i.e. to solve

$$\frac{\partial \chi^2}{\partial \theta_j} = 0$$

Although it is not obvious yet, it turns out that dividing the residuals by σ_i (as opposed to σ_i^2 or some other function) gives a good estimation method if the y_i have Gaussian distributions and this is where this assumption is implicitly needed.

Note, if all the uncertainties on the points are the same, i.e. $\sigma_i = \sigma$, then the chi-squared becomes

$$\chi^2(\theta_j; y_i) = \sum_{i=1}^N \left[\frac{y_i - f(x_i; \theta_j)}{\sigma} \right]^2 = \frac{1}{\sigma^2} \sum_{i=1}^N [y_i - f(x_i; \theta_j)]^2 = \frac{S}{\sigma^2}$$

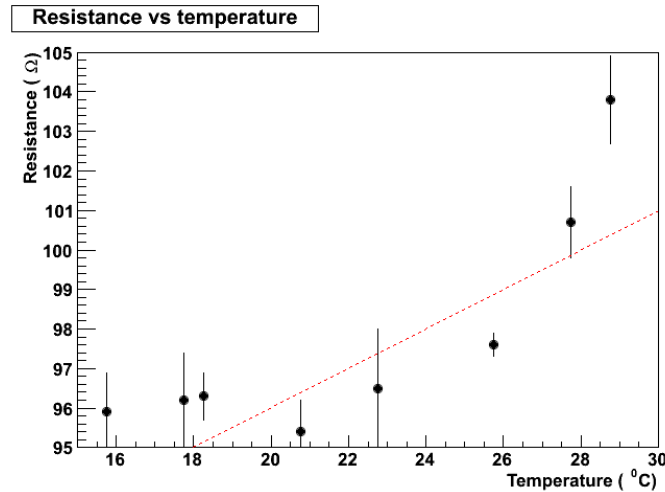
Hence the chi-squared is equal to the sum of squares of the residuals S multiplied by a constant factor. This means the values of the parameters θ_j which minimise the chi-squared will be identical to those resulting from the least squares method. Hence, the estimates of the parameters from the two methods will be the same in this particular case.

Solving for the minimum can be done analytically for simple functions, but in most cases must be done numerically on a computer. Such a process of finding the values of parameters which minimise (or maximise) a function is generally called ‘fitting’ the parameters. Both the chi-squared and the maximum likelihood methods are therefore examples of fitting the parameters.

16 Goodness of fit

Besides allowing for differing sizes of uncertainties on the y_i , the chi-squared method is more useful even if all the σ_i have the same value (when it gives identical estimates for the θ_j as the simple least squares method). This is because the minimum chi-squared value itself gives further information, specifically on what is called the “goodness of fit”. This tells us if the function we have assumed is a sensible one or not.

It is clear that if the function *is* correct, then the values of the y_i and the function $f(x_i; \theta_j)$ for each x_i should not be too different. Conversely, a bad fit will have the points far from the function; see an example below.



Specifically, for a good fit, we would expect the values of the y_i to differ from the fitted function by amounts similar to the σ_i as they are (assumed to be) Gaussian distributed. Specifically, if the fitted function is sensible, it should be close to the mean for that x_i and hence the residual (i.e. the fluctuation around that mean) would be expected to have a standard deviation around the fitted function of roughly σ_i . Hence, each term in the chi-squared sum would be expected to be $(r_i/\sigma_i)^2 \sim 1$ and hence the chi-squared at the minimum should have a value of $\chi^2_{\min} \sim N_{\text{data}}$. A much bigger value would imply that the wrong function is being used and so can help to identify if an underlying theory is incorrect.

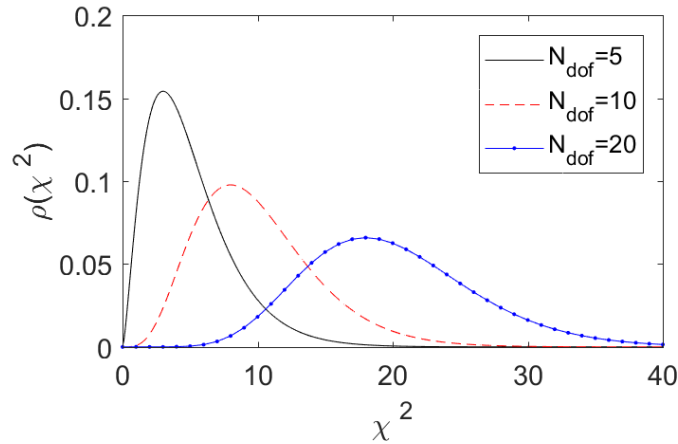
To be more precise, it is not actually correct to compare the chi-squared value to N_{data} , but a correction is needed. To see this, consider an extreme example; a straight line fit to two data

points will always give a minimum chi-squared of exactly zero. This does not mean the straight line fit is good; this is just saying there is a solution, not a fit, if the number of data points is equal to the number of parameters. It turns out the critical measure is the “number of degrees of freedom” N_{dof} , which is defined to be the difference between the number of data points being used in the fit and the number of parameters being estimated in the fit

$$N_{\text{dof}} = N_{\text{data}} - N_{\text{paras}}$$

If there are more data points than parameters, i.e. $N_{\text{dof}} > 0$, then there are more constraints than there are parameters to be determined and a fit is needed. The χ^2_{min} value then gives information on the goodness of fit. If the number of data points is equal to the number of parameters, i.e. $N_{\text{dof}} = 0$, then the equations can be solved rather than fitted, and the chi-squared will be identically zero so we have no information on the goodness of fit. Finally, if the number of data points is less than the number of parameters, i.e. $N_{\text{dof}} < 0$, the system is underconstrained and no unique determination of the parameters is possible. In terms of the minimum value of the chi-squared, it turns out that for $N_{\text{dof}} > 0$ we would expect $\chi^2_{\text{min}} \sim N_{\text{dof}}$ for a good fit, while values much bigger than this would imply the function being fitted does not actually describe the data well.

The chi-squared is actually a random variable with a PDF. At a basic level, given that all the measurement distributions are Gaussian, then we can calculate this PDF for the chi-squared value. It turns out this only depends on the number of degrees of freedom, not the number of data points or parameters separately, i.e. $\rho(\chi^2; N_{\text{dof}})$. The PDF has an expectation value of N_{dof} and has a long tail out to infinity, as shown in the plot below for different N_{dof} .



The goodness of fit is therefore effectively a statement about whether our fit gives us a chi-squared value near the PDF peak or in the tail at higher chi-squared. This is often quantified in terms of a “p-value” which gives the probability of getting the observed or a worse (higher) chi-squared value. There are tables of p-values in statistics books and on the web, and software packages may have functions to calculate the p-value from a given chi-squared and number of degrees of freedom. However, as the p-value corresponds to integrating the chi-squared distribution from the observed value up to infinity, then it corresponds exactly to what we previously called the one-sided significance level, set by the data value. Hence, stating that the goodness of fit is poor is precisely a hypothesis test, i.e. we say we reject the hypothesis that the data agree with the function chosen for the fit if the goodness of fit is bad.

17 Chi-squared method for binned distributions

We are often trying to find the parameters of a function which describes the shape of some data distribution (e.g. the shape of the histogram). By separating the data into small ranges (“bins”) as in a histogram, we can count the number of times the variable falls into each bin.

This lends itself to a chi-squared treatment if the numbers of events in each bin are large, i.e. $n_i \gg 1$ for each bin i . This is because the probability distribution for the number n_i in a single bin is (to a good approximation) a Poisson, where the mean of the Poisson is just the integral of the function over the bin. Specifically

$$\mu_i(\theta_j) = \int_{\text{Bin } i} f(x; \theta_j) dx$$

It is also common in practise that μ_i is approximated to

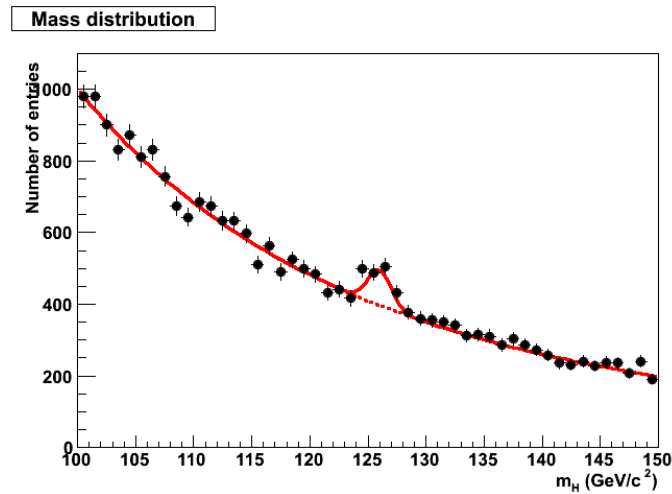
$$\mu_i(\theta_j) \approx f(x_i; \theta_j) \Delta x$$

where x_i is the value of x at the centre of each bin and Δx is the bin width. Given this, n_i has a distribution of $P[n_i; \mu_i(\theta_j)]$. Hence, because of the Central Limit Theorem for large n_i as seen in Lecture 4, this is approximately Gaussian, with $\sigma_i \approx \sqrt{n_i}$. Hence the chi-squared can be approximated by

$$\chi^2(\theta_j; \underline{n}) = \sum_{i=1}^N \frac{[n_i - \mu_i(\theta_j)]^2}{n_i}$$

where the sum is over all the bins. This method is widely used but it is important it is only done when all bins have a large number of entries, i.e. $n_i \gg 1$, which is typically at least 20 entries.

An example of this application is the discovery of the Higgs boson; it was discovered in 2012 by finding the mass of its decay products. There is a large background of random combinations which gave a smooth shape in the mass distribution, while the Higgs itself gives a narrow Gaussian peak in mass. By fitting for the background shape parameters, the background-only hypothesis could be rejected due to a poor goodness of fit; in the Higgs case the significance level was 5σ which is very strong evidence for rejection. Knowing that the smooth background function was not sufficient for a good fit to the data, including parameters for the mean (i.e. the Higgs mass) and width of the Gaussian allowed its existence to be established and its mass to be determined.



18 Connection to maximum likelihood estimation

Let's consider the case of a set of measurements y_i of Gaussian distributed random variables, each with a different mean μ_i and width σ_i . We will assume the means depend on some parameters θ_j so $\mu_i(\theta_j)$, but that the widths σ_i are known. The likelihood for one such measurement is

$$L_i = \frac{1}{\sigma_i \sqrt{2\pi}} e^{-(y_i - \mu_i)^2 / 2\sigma_i^2}$$

so the log-likelihood is

$$\ln(L_i) = -\frac{(y_i - \mu_i)^2}{2\sigma_i^2} - \ln(\sigma_i \sqrt{2\pi})$$

Hence, for all the measurements, the total log-likelihood is

$$\ln(L) = \sum_i \ln(L_i) = -\sum_i \frac{(y_i - \mu_i)^2}{2\sigma_i^2} - \sum_i \ln(\sigma_i \sqrt{2\pi}) = -\sum_i \frac{(y_i - \mu_i)^2}{2\sigma_i^2} + C$$

where C is a term which does not depend on the μ_i and hence also not on the θ_j parameters. This means

$$2C - 2\ln(L) = \sum_i \frac{(y_i - \mu_i)^2}{\sigma_i^2}$$

The term on the right can be seen to be what we defined as the chi-squared, i.e.

$$\chi^2 = 2C - 2\ln(L)$$

When we considered the chi-squared, we discussed the values as being described by some function but effectively all this does is to define a different mean for each measurement, i.e.

$$\mu_i(\theta_j) = f(x_i; \theta_j)$$

In these terms, then we would maximise the log-likelihood to find the estimates $\hat{\theta}_j$. However, since they differ by a sign, *maximising* the log-likelihood is equivalent to *minimising* the chi-squared and hence the estimates from either method will be identical. In fact, it is common to work with $-2\ln(L)$ and find a minimum rather than a maximum, as this is then directly comparable to the chi-squared in the Gaussian case.

In general however, there is no goodness-of-fit measure for a likelihood fit, unlike the chi-squared method. In the Gaussian case, the constant C above is calculable and so the chi-squared can be found from the maximum likelihood. However, for non-Gaussian cases (which are exactly when the maximum likelihood is most useful), the above form is not found and there is no such goodness-of-fit measure.

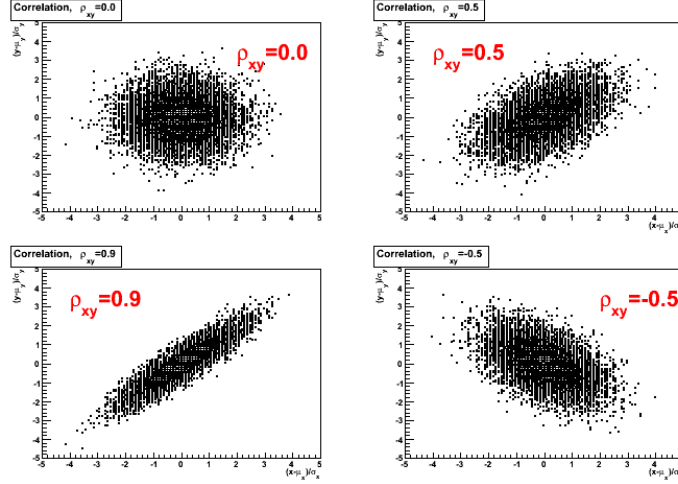
19 Non-examinable: the error matrix

The general formulation for uncertainties with more than one variable is using the 'error matrix' $\underline{\underline{E}}$, sometimes called the 'covariance matrix'. For independent (so-called 'uncorrelated') variables, the error matrix is diagonal and equal to

$$\underline{\underline{E}} = \begin{pmatrix} \sigma_1^2 & 0 & 0 & \dots \\ 0 & \sigma_2^2 & 0 & \dots \\ 0 & 0 & \sigma_3^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where σ_i is the uncertainty on variable i .

This can be generalised; the zeros off-diagonal indicate that there are no correlations between the measurements. However, conversely, if the variables *are* correlated, then these off-diagonal elements are no longer zero, but have the form $\rho_{ij}\sigma_i\sigma_j$, where ρ_{ij} , the ‘correlation coefficient’, must be in the range $-1 \leq \rho_{ij} \leq 1$. Clearly, $\rho_{ij} = 0$ corresponds to the uncorrelated case. For non-zero ρ_{ij} , then if one variable fluctuates higher than the average, then the other will also tend to fluctuate higher (if they are correlated, i.e. $\rho_{ij} > 0$) or lower (if they are anticorrelated, i.e. $\rho_{ij} < 0$), as shown in the figure below.



Hence, the general error matrix is

$$\underline{\underline{E}} = \begin{pmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 & \dots \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 & \dots \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Knowing about the error matrix allows us to write down the most general form for the propagation of errors formula. If a quantity z depends on several variables x_i , i.e. $z = z(\underline{x})$, then by forming a ‘derivative vector’ $\underline{d}$

$$\underline{d} = \begin{pmatrix} \partial z / \partial x_1 \\ \partial z / \partial x_2 \\ \partial z / \partial x_3 \\ \vdots \end{pmatrix}$$

the error on z is given by

$$\sigma_z^2 = \underline{d}^T \underline{\underline{E}} \underline{d}$$

For example, with two variables $x_1 = x$, $x_2 = y$, then generally

$$\begin{aligned} \sigma_z^2 &= \begin{pmatrix} \partial z / \partial x & \partial z / \partial y \end{pmatrix} \begin{bmatrix} \sigma_x^2 & \rho_{xy}\sigma_x\sigma_y \\ \rho_{xy}\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} \begin{pmatrix} \partial z / \partial x \\ \partial z / \partial y \end{pmatrix} \\ &= \begin{pmatrix} \partial z / \partial x & \partial z / \partial y \end{pmatrix} \begin{bmatrix} (\partial z / \partial x)\sigma_x^2 + (\partial z / \partial y)\rho_{xy}\sigma_x\sigma_y \\ (\partial z / \partial x)\rho_{xy}\sigma_x\sigma_y + (\partial z / \partial y)\sigma_y^2 \end{bmatrix} \\ &= \left(\frac{\partial z}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial z}{\partial y}\right)^2 \sigma_y^2 + 2\left(\frac{\partial z}{\partial x}\right)\left(\frac{\partial z}{\partial y}\right)\rho_{xy}\sigma_x\sigma_y \end{aligned}$$

This reduces to the form given previously for the propagation of errors formula in the uncorrelated case, i.e. when $\rho_{xy} = 0$.

20 Non-examinable: chi-squared method for correlated uncertainties

There is an even more complete version of the chi-squared method, which occurs when the measurements y_i are correlated. The general formulation for uncertainties with more than one variable is using the error matrix $\underline{\underline{E}}$, as discussed in the previous section.

The expression for the chi-squared can be written in terms of vectors and matrices. Defining the “residual vector” $\underline{r}$ as

$$\underline{r} = \begin{bmatrix} y_1 - f(x_1; \theta_j) \\ y_2 - f(x_2; \theta_j) \\ y_3 - f(x_3; \theta_j) \\ \vdots \end{bmatrix}$$

so its transpose is

$$\underline{r}^T = [y_1 - f(x_1; \theta_j) \quad y_2 - f(x_2; \theta_j) \quad y_3 - f(x_3; \theta_j) \quad \dots]$$

then the chi-squared can be written as

$$\chi^2 = [y_1 - f(x_1; \theta_j) \quad y_2 - f(x_2; \theta_j) \quad y_3 - f(x_3; \theta_j) \quad \dots] \begin{pmatrix} 1/\sigma_1^2 & 0 & 0 & \dots \\ 0 & 1/\sigma_2^2 & 0 & \dots \\ 0 & 0 & 1/\sigma_3^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{bmatrix} y_1 - f(x_1; \theta_j) \\ y_2 - f(x_2; \theta_j) \\ y_3 - f(x_3; \theta_j) \\ \vdots \end{bmatrix}$$

The matrix in the middle is called the “weight matrix” $\underline{\underline{W}}$ and the chi-squared can be written compactly as

$$\chi^2 = \underline{r}^T \underline{\underline{W}} \underline{r}$$

The inverse of the weight matrix is obviously

$$\underline{\underline{E}} = \underline{\underline{W}}^{-1} = \begin{pmatrix} \sigma_1^2 & 0 & 0 & \dots \\ 0 & \sigma_2^2 & 0 & \dots \\ 0 & 0 & \sigma_3^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

From above, this will be recognised as the form of the error matrix for uncorrelated variables. As discussed previously, the more general error matrix, allowing for possible correlations, is

$$\underline{\underline{E}} = \begin{pmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 & \dots \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 & \dots \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Inverting this matrix to get the weight matrix then allows the chi-squared to be expressed in the same way as before

$$\chi^2 = \underline{r}^T \underline{\underline{W}} \underline{r}$$

This is the general form; even for the correlated case, then combining the residual vectors and weight matrix as above gives the chi-squared.

Statistics of Measurement

Lecture 8 - Confidence intervals and parameter uncertainties

Heather Graven, 25 May 2023

21 Introduction

We have discussed estimating parameters in the last two lectures, using either the maximum likelihood or chi-squared methods. However, we have not discussed how to find the uncertainty on the parameters we estimate. We have also not defined precisely what we mean by an uncertainty on a parameter. This is where the real differences between frequentist and Bayesian definitions of probability arise. We shall discuss the frequentist approach here and then discuss Bayesian estimation in the next lecture.

The first thing to do is understand how to find the uncertainties on the parameters when using the two methods we have studied. In principle there is nothing new here. Both methods can be considered as giving an equation for the parameter(s) θ as a function of the measurements x_i

$$\hat{\theta} = M_{\theta}(x_1, x_2, \dots, x_N)$$

so, at least in a linear approximation, we can just apply the propagation of errors formula to get the uncertainty on each θ_i , i.e.

$$\sigma_{\theta_i}^2 = \sum_i \left(\frac{\partial M_{\theta}}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

However, in practical terms this can often be inconvenient to calculate, particularly when the maximum or minimum is only found numerically. Also, the propagation of errors equation is only itself an approximation. Let's consider each of the two methods in turn.

22 Maximum likelihood parameter uncertainties

The central limit theorem says for large N , the mean and variance of the sum of repeated measurements of random variables are distributed according to a Gaussian of mean $\mu = \sum_i E_i$ and width $\sigma^2 = \sum_i V_i$. With enough measurements, many PDFs approximate to Gaussians. It turns out that the likelihood's dependence on the parameters is then also Gaussian. It is therefore very common to take an approximation that the likelihood is close to a Gaussian.

For simplicity, assume there is only one parameter θ . We can do a Taylor expansion of the log-likelihood in terms of θ around the maximum, which is where the parameter is $\theta = \hat{\theta}$. The Taylor expansion then gives

$$\ln[L(\theta)] = \ln[L(\hat{\theta})] + \left. \frac{d \ln(L)}{d\theta} \right|_{\hat{\theta}} (\theta - \hat{\theta}) + \left. \frac{d^2 \ln(L)}{d\theta^2} \right|_{\hat{\theta}} \frac{(\theta - \hat{\theta})^2}{2!} + \dots$$

Because $\hat{\theta}$ is defined to be at the maximum, the first derivative is zero. The second derivative is negative (since it is a maximum) but is some constant when evaluated at $\hat{\theta}$; let this be

$$\frac{1}{\Sigma^2} = - \left. \frac{d^2 \ln(L)}{d\theta^2} \right|_{\hat{\theta}}$$

where the negative sign means Σ^2 is positive and hence Σ is real. In terms of Σ , then

$$\ln[L(\theta)] = \ln[L(\hat{\theta})] - \frac{(\theta - \hat{\theta})^2}{2\Sigma^2} + \dots$$

If the higher order terms are now ignored, then taking exponentials gives

$$L(\theta) \approx L(\hat{\theta})e^{-(\theta-\hat{\theta})^2/2\Sigma^2}$$

which is clearly a Gaussian function for θ . It is also seen that the quantity Σ generally is the width of the Gaussian, which is therefore the uncertainty on $\hat{\theta}$. Hence, in this approximation (or exactly if the function is truly a Gaussian), then the uncertainty is given by the equation above involving the second derivative, evaluated with $\theta = \hat{\theta}$. Warning: the above equation for $1/\Sigma^2$ is only valid for the one parameter case. For more parameters, this gives the inverse of a matrix, which has to be inverted to get the uncertainties (see the appendix).

Example: We have a large number N of measurements of an exponential random variable. We will express the exponential PDF as $e^{-x/a}/a$, where the parameter a (to be estimated) is the average, as shown in Lecture 3. The likelihood and hence log-likelihood for a single measurement is

$$L_i(a) = \frac{1}{a}e^{-x_i/a} \quad \text{so} \quad \ln[L_i(a)] = -\ln(a) - \frac{x_i}{a}$$

Hence, the total log-likelihood is

$$\ln[L(a)] = \sum_i \ln[L_i(a)] = -N \ln(a) - \frac{1}{a} \sum_i x_i$$

The derivative is

$$\frac{d \ln(L)}{da} = -\frac{N}{a} + \frac{1}{a^2} \sum_i x_i$$

so the estimate is given by this being zero, for which

$$\frac{N}{\hat{a}} = \frac{1}{\hat{a}^2} \sum_i x_i \quad \text{so} \quad \hat{a} = \sum_i x_i / N$$

as would be expected. The second derivative is

$$\frac{d^2 \ln(L)}{da^2} = \frac{N}{a^2} - \frac{2}{a^3} \sum_i x_i \quad \text{so} \quad \left. \frac{d^2 \ln(L)}{da^2} \right|_{\hat{a}} = -\frac{N}{\hat{a}^2}$$

and so is negative as required for a maximum. Therefore, the uncertainty on the estimate is approximately

$$\sigma_{\hat{a}} \approx \frac{1}{\sqrt{-d^2 \ln(L)/da^2|_{\hat{a}}}} = \frac{\hat{a}}{\sqrt{N}}$$

The same result would be obtained from using the propagation of errors formula on the result for $\hat{a}$, remembering that the standard deviation of an exponential random variable is a . Explicitly

$$\frac{\partial \hat{a}}{\partial x_j} = \frac{1}{N}$$

so

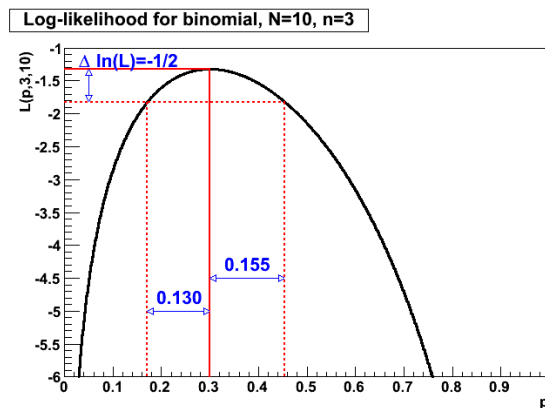
$$\sigma_{\hat{a}}^2 = \sum_j \left(\frac{\partial \hat{a}}{\partial x_j} \right)^2 \sigma_{x_j}^2 = \sum_j \frac{1}{N^2} \hat{a}^2 = \frac{\hat{a}^2}{N}$$

giving the same result for $\sigma_{\hat{a}}$.

What if the PDF is not even approximately Gaussian or/and propagation of errors is not a good approximation? The uncertainty can still be evaluated but by using a different method. For the Gaussian case, then the value of the log-likelihood at $\theta = \hat{\theta} \pm \Sigma$ is

$$\ln[L(\theta)] = \ln[L(\hat{\theta})] - \frac{(\hat{\theta} \pm \Sigma - \hat{\theta})^2}{2\Sigma^2} = \ln[L(\hat{\theta})] - \frac{\Sigma^2}{2\Sigma^2} = \ln[L(\hat{\theta})] - \frac{1}{2}$$

Hence, the uncertainty is the range which makes the log-likelihood change by $-1/2$. It turns out this is true for non-Gaussian likelihoods too; the uncertainty can be evaluated by changing the parameter and seeing what range gives $\Delta \ln(L) \geq -1/2$. (In principle, this is only exact in the large N limit, but it is normally a very good approximation anyway.) In general, this can require a different shift in θ when going up compared to down, and so this gives asymmetric uncertainties. An example for a binomial measurement is shown below. In this case, you will see them written as e.g. $0.30^{+0.15}_{-0.13}$ for this example.



23 Chi-squared parameter uncertainties

As discussed in lecture 7, the chi-squared is a function of the parameters

$$\chi^2(\theta; y_i) = \sum_{i=1}^N \left[\frac{y_i - f(x_i; \theta)}{\sigma_i} \right]^2$$

and the estimates of the parameters are taken to be the values that give the minimum chi-squared.

However, consider expanding the chi-squared around the minimum using a Taylor expansion in the parameters. We will again consider only one parameter for clarity. Since it is a minimum, there is no linear term and so we know immediately it must be of the form

$$\chi^2(\theta) \approx \chi^2(\hat{\theta}) + \left. \frac{d^2 \chi^2}{d\theta^2} \right|_{\hat{\theta}} \frac{(\theta - \hat{\theta})^2}{2} + \dots$$

Since the best-fit chi-squared is a minimum, then the second derivative must be positive. Hence, let us express the second derivative as

$$\left. \frac{d^2 \chi^2}{d\theta^2} \right|_{\hat{\theta}} = \frac{2}{\Sigma^2} \quad \text{i.e.} \quad \frac{1}{\Sigma^2} = \frac{1}{2} \left. \frac{d^2 \chi^2}{d\theta^2} \right|_{\hat{\theta}}$$

such that we can write

$$\chi^2(\theta) \approx \chi^2(\hat{\theta}) + \frac{(\theta - \hat{\theta})^2}{\Sigma^2}$$

We can interpret this as being some sort of ‘residual’ of the parameters around the best estimate of $\hat{\theta}$. With this interpretation, then it is clear the uncertainty on the best estimate is Σ . It can be

shown (see appendix) that Σ is indeed the uncertainty on the parameter that would be obtained from using the propagation of errors formula, i.e.

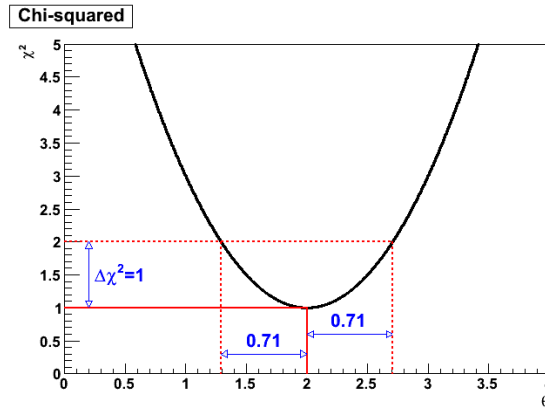
$$\Sigma^2 = \sum_i \left(\frac{\partial \theta}{\partial y_i} \right)^2 \sigma_i^2$$

so that our interpretation is indeed correct.

It is sometimes the case that the second derivative is easy to calculate and so the uncertainty can be found using the above method. However, the chi-squared method can be used for arbitrarily complicated functions $f(x_i; \theta)$ and in many cases it is impracticable to find the uncertainty this way. Luckily, there is a simple numerical method, which is to change the parameter from its best estimate and see what happens to the value of the chi-squared. If we shift the parameter θ from $\hat{\theta}$ (the value which gives the minimum chi-squared) to $\hat{\theta} \pm \Sigma$, then the chi-squared becomes

$$\chi^2(\hat{\theta} \pm \Sigma) \approx \chi^2(\hat{\theta}) + \frac{\Sigma^2}{\Sigma^2} \approx \chi^2(\hat{\theta}) + 1$$

Hence, if we numerically evaluate the chi-squared for various different values of θ , then the range for which the chi-squared changes by up to one unit gives the uncertainty on $\hat{\theta}$. An example is shown below, where the result would be quoted as $\hat{\theta} = 2.0 \pm 0.7$.



We showed in the last lecture that for the Gaussian case, the chi-squared is related to the log-likelihood by $\chi^2 \sim -2 \ln(L)$. Hence, changing the chi-squared by +1 is equivalent to changing the log-likelihood by $-1/2$ and so these two uncertainty estimates are in agreement.

24 Confidence intervals for random variables

Let's think about confidence intervals in relation to random variables, then we will go back to estimated parameters in the next section. Back in lecture 3, we discussed the probability contained within certain ranges of the Gaussian PDF, e.g. the region $\pm 1\sigma$ of the mean contained 68.3% of the integrated PDF

$$\int_{\mu-\sigma}^{\mu+\sigma} G(x; \mu, \sigma) dx = \frac{2}{\sqrt{\pi}} \int_0^{1/\sqrt{2}} e^{-y^2} dy = \text{erf}(1/\sqrt{2}) = 0.683$$

The same could be done for other ranges; e.g. the probability within $\pm 2\sigma$ is

$$\int_{\mu-2\sigma}^{\mu+2\sigma} G(x; \mu, \sigma) dx = \frac{2}{\sqrt{\pi}} \int_0^{2/\sqrt{2}} e^{-y^2} dy = \text{erf}(2/\sqrt{2}) = 0.954$$

Considering the $\pm 1\sigma$ case, this means that a random measurement of x arising from a Gaussian distribution has a probability of 68.3% of lying within this range. This is referred to as the ‘confidence interval’, i.e. we have 68.3% confidence that the value will lie in the interval within $\pm 1\sigma$ of the mean. Hence, if we take many such measurements, then 68.3% of them (which is roughly 2/3) will be within $\pm 1\sigma$ of the mean.

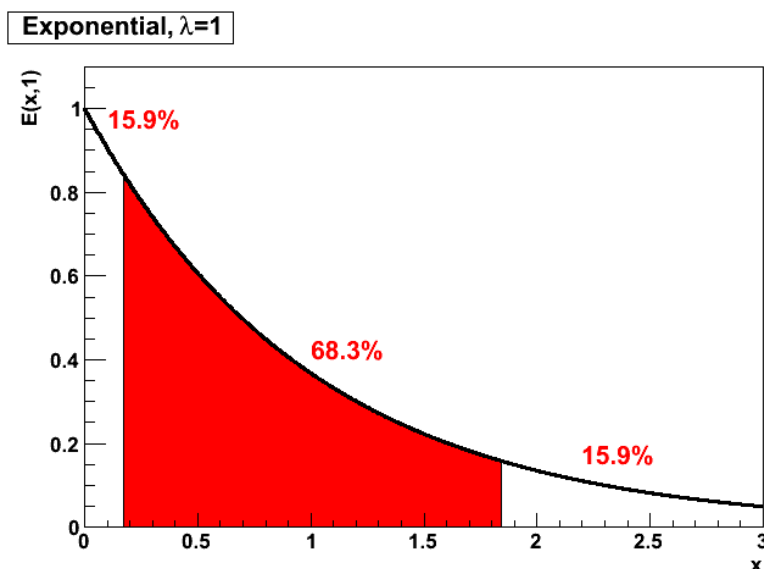
Ranges can also be defined for non-Gaussian distributions. Even though the specific value of 68.3% arises from the integration of the Gaussian, it is still very common to use a range containing this amount of the integral, even for non-Gaussian distributions. It is even common to refer to this 68.3% confidence range as being the ‘one sigma’ range, even though there may be no such parameter σ in the distribution, and the standard deviation may represent a different fraction of the total probability. However, even in trying to calculate the 68.3% range, there is complication, namely non-Gaussian distributions can be asymmetric. In this case, there is no unique answer as to how to define this range. We need

$$\int_a^b \rho(x) dx = 0.683$$

but this only gives one constraint on the two limits a and b . For example, for an exponential

$$\int_a^b \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_a^b = e^{-\lambda a} - e^{-\lambda b} = 0.683$$

we have two unknowns in the equation. A common way to handle this is to say that the amount outside the range, which is $100\% - 68.3\% = 31.7\%$, has to be equally divided above and below the range, so there is 15.9% on either side. This is shown for an exponential below.



25 Confidence intervals for estimated parameters

The meaning of confidence intervals for random variables should hopefully be clear. The confidence interval is associated with the specific probability of the random variable being within that interval, for example, the 68.3% confidence interval for a random variable with a Gaussian PDF is $\pm 1\sigma$.

Now back to parameters. Parameters are *not* random variables but have a exact value, even if this is unknown to us. E.g. the electron mass is a fixed value, even if we are trying to measure it in an experiment.

So what do we really mean by uncertainties on parameters, e.g. by finding $\Delta \ln(L) = -1/2$ or $\Delta \chi^2 = +1$? What we are in fact doing is giving a range. Clearly, the true value is either within this range or is not, and so any particular experiment may be right or wrong about the true value being within the range. However, the range chosen is such that if we did a large number of experiments to find the parameter value, then 68.3% (or 95% or whatever the quoted confidence interval) of those experiments will contain the true value within the derived confidence interval. This is the meaning of a confidence interval for a parameter in the frequentist interpretation and this is what we actually mean when we quote the ‘uncertainty’ on a parameter. This is subtly different from that for a random variable and can take a while to understand. In practise, we handle the parameter errors in the same way as random variable errors, e.g. using the propagation of errors formula, but there is an underlying conceptual difference.

Furthermore, sometimes we are only interested in one side of the confidence interval, e.g. if we are trying to determine a rate of a very rare process, we know a rate cannot be negative so we only want to set an upper limit. In this case, it is common to use the 90% or, more usually, 95% confidence level, but only on one side of the range. Hence, just like hypothesis testing, we can set one-sided or two-sided confidence intervals.

26 Physical constraints

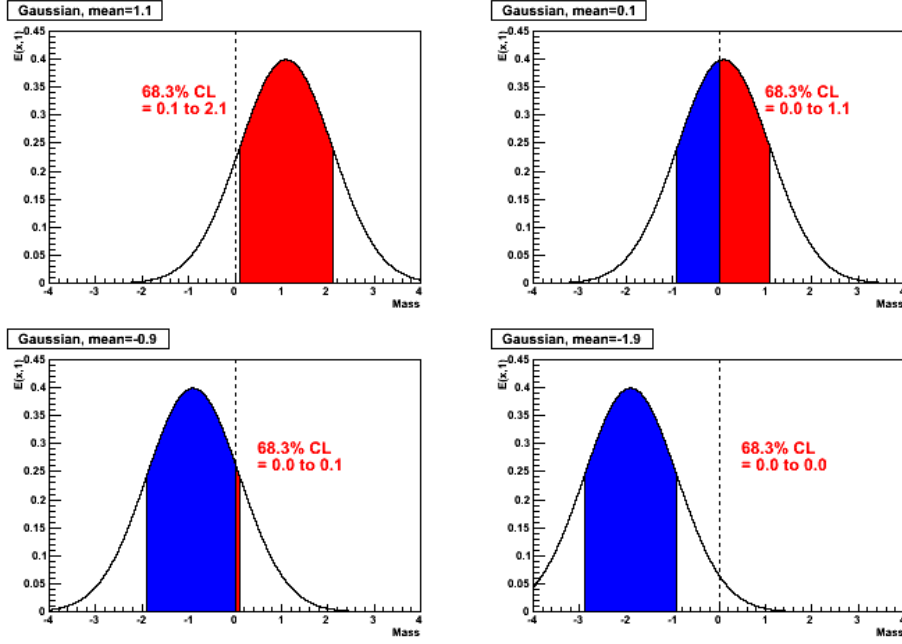
One of the problems with frequentist methods is that we can end up with results outside the physical region. For example, we want to measure the mass of some powder of around 1 g using a set of weighing scales with 1% accuracy. The dish containing the powder is known to weigh 100 g so the uncertainty on the weight is 1 g. If we happened to measure 101.1 g then in the frequentist approach, we would set the 68.3% confidence interval for the mass of the powder to be 0.1 to 2.1 g, which is fine. However, it is not unlikely that the measurement could give something like 100.1 g, which would give a range from -0.9 to 1.1 g. However, the mass cannot be negative so the 68.3% confidence interval would have to be quoted as 0.0 to 1.1 g, which looks like it is more accurate, even though it uses the same apparatus. A bigger problem arises if the measurement happens to yield 99.1 g; the range would be -1.9 to 0.1 g so the apparent confidence interval would be 0.0 to 0.1 g. However, saying the mass is below 0.1 g when the scales are only accurate to 1 g is clearly not right. A final example might be a fluctuation where the measurement yields 98.1 g; the range would be -2.9 to -0.9 g which means there is no 68.3% confidence interval in the physical region. These cases are illustrated in the plots below.

We know these low measurements must have been ‘unlucky’ in some sense. Hence, we would expect them to be more likely to be some of the 31.7% of confidence intervals which do *not* contain the true value. However, it is very hard to quantify this in a consistent way in the frequentist approach. We will see how the Bayesian approach handles this in the next lecture.

27 Non-examinable: Appendix

27.1 Single parameter uncertainty from chi-squared

This section goes through a derivation of the uncertainty of a single parameter and compares it with a propagation of errors approach like we did before with the likelihood, but now using χ^2 . Consider a simple chi-squared case with one parameter θ , where the function is linear in the



parameter (although it can be arbitrarily complicated in x), so

$$f(x; \theta) = a(x) + b(x)\theta$$

for any functions $a(x)$ and $b(x)$. Hence,

$$\chi^2(\underline{\theta}; \underline{y}) = \sum_{i=1}^N \left[\frac{y_i - f(x_i; \theta)}{\sigma_i} \right]^2 = \sum_{i=1}^N \left[\frac{y_i - a(x_i) - b(x_i)\theta}{\sigma_i} \right]^2$$

and the estimate of the parameter is given by solving

$$\left. \frac{d\chi^2}{d\theta} \right|_{\hat{\theta}} = 0 = -2 \sum_{i=1}^N \frac{[y_i - a(x_i) - b(x_i)\hat{\theta}]b(x_i)}{\sigma_i^2}$$

This gives

$$\sum_{i=1}^N \frac{[y_i - a(x_i)]b(x_i)}{\sigma_i^2} = \hat{\theta} \sum_{j=1}^N \frac{b(x_j)^2}{\sigma_j^2}$$

so

$$\hat{\theta} = \frac{\sum_{i=1}^N [y_i - a(x_i)]b(x_i)/\sigma_i^2}{\sum_{j=1}^N b(x_j)^2/\sigma_j^2}$$

Note, the σ_i^2 terms do *not* cancel out here as they are inside separate summations.

The uncertainty on $\hat{\theta}$ arises as the y_i have uncertainties σ_i . Hence, we can use propagation of errors to find this uncertainty. From above

$$\frac{\partial \hat{\theta}}{\partial y_i} = \frac{b(x_i)/\sigma_i^2}{\sum_{j=1}^N b(x_j)^2/\sigma_j^2}$$

so that by propagation of errors

$$\sigma_{\hat{\theta}}^2 = \sum_{i=1}^N \left(\frac{\partial \hat{\theta}}{\partial y_i} \right)^2 \sigma_i^2 = \sum_{i=1}^N \frac{b(x_i)^2/\sigma_i^4}{\left(\sum_{j=1}^N b(x_j)^2/\sigma_j^2 \right)^2} \sigma_i^2 = \frac{\sum_{i=1}^N b(x_i)^2/\sigma_i^2}{\left(\sum_{j=1}^N b(x_j)^2/\sigma_j^2 \right)^2} = \frac{1}{\sum_{i=1}^N b(x_i)^2/\sigma_i^2}$$

However, it is claimed in the main part of the lecture that this is equal to the uncertainty Σ given by the second derivative

$$\frac{1}{\Sigma^2} = \frac{1}{2} \left. \frac{d^2 \chi^2}{d\theta^2} \right|_{\hat{\theta}}$$

The second derivative is

$$\frac{d^2 \chi^2}{d\theta^2} = 2 \sum_{i=1}^N \frac{b(x_i)^2}{\sigma_i^2} = \frac{2}{\sigma_{\hat{\theta}}^2}$$

and so $\Sigma = \sigma_{\hat{\theta}}$ as claimed. Effectively any function can be approximated to being linear in its parameter by doing a Taylor expansion around the estimate, so the above holds approximately in general, with the approximation being good for small uncertainties.

This means that for the chi-squared, and for the maximum likelihood in the Gaussian approximation, the uncertainty on a single parameter is given by either

$$\frac{1}{\sigma_{\hat{\theta}}^2} = \frac{1}{2} \left. \frac{d^2 \chi^2}{d\theta^2} \right|_{\hat{\theta}} \quad \text{or} \quad \frac{1}{\sigma_{\hat{\theta}}^2} = - \left. \frac{d^2 \ln(L)}{d\theta^2} \right|_{\hat{\theta}}$$

respectively.

27.2 Error matrix for more than one parameter

In the Gaussian approximation for more than one parameter, then we have to use the error matrix $\underline{\underline{E}}$, or more specifically the weight matrix $\underline{\underline{W}} = \underline{\underline{E}}^{-1}$ mentioned in the notes for Lecture 7. The generalisation of the above expressions is in terms of the elements of the weight matrix W_{ij} , where

$$W_{ij} = \frac{1}{2} \left. \frac{\partial^2 \chi^2}{\partial \theta_i \partial \theta_j} \right|_{\hat{\theta}} \quad \text{or} \quad W_{ij} = - \left. \frac{\partial^2 \ln(L)}{\partial \theta_i \partial \theta_j} \right|_{\hat{\theta}}$$

Statistics of Measurement

Lecture 9 - Bayesian estimation

Heather Graven, 26 May 2023

28 Introduction

We saw in the previous lecture that a frequentist treatment of parameter estimates interprets them differently from random variables. As we will see in this lecture, the Bayesian approach allows us to handle these two cases in the same fashion. Sometimes this is at a cost of introducing a subjective element.

29 Bayesian probabilities

As mentioned briefly in Lecture 1, the Bayesian concept of a probability is subjective and depends on the available information. There is no need in the Bayesian framework to define multiple experiments which are repeatable. Hence, the Bayesian concept can be applied more widely.

In particular, in the Bayesian framework we can consider the parameters determined by our experiments as probabilistic variables. We are not saying e.g. that the true electron mass is not a precise value; what we mean is that the probability we assign to its possible values reflects the degree of our (lack of) knowledge. While this seems a minor change of philosophy, it means we can apply all the probability theory we have already learned for random variables directly to the parameter also. In particular, once we have the parameter probability distribution, we can find our 68.3% confidence interval (and hence uncertainties) just as we did for random variables.

In particular, we can apply Bayes' theorem of conditional probabilities to the parameters as well as the measurements. We saw Bayes' theorem in Lecture 1

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

where $P(x|y)$ is the probability for x given the value of y . Now, if we say x corresponds to the results from the experiment we have performed (the data) and y corresponds to the parameter θ , then this reads

$$P(\theta|\text{Data}) = \frac{P(\text{Data}|\theta)P(\theta)}{P(\text{Data})}$$

This allows us to calculate the conditional probability of the parameters given the data we have measured (the left-hand side) from the conditional probability of the data given the parameters (the numerator on the right-hand side). This term is just equivalent to the likelihood, which we met in Lecture 6. Multiplied by the likelihood is $P(\theta)$, the probability for the parameter, independent of the outcome of the experiment; i.e. before the experiment is done. (This is sometimes difficult to evaluate and this issue is discussed further below.) Finally, the denominator is the probability of seeing the data we measured, independent of any particular values of the parameters. This is sometimes called the 'evidence' but it effectively acts as a normalisation constant, i.e. for a discrete parameter

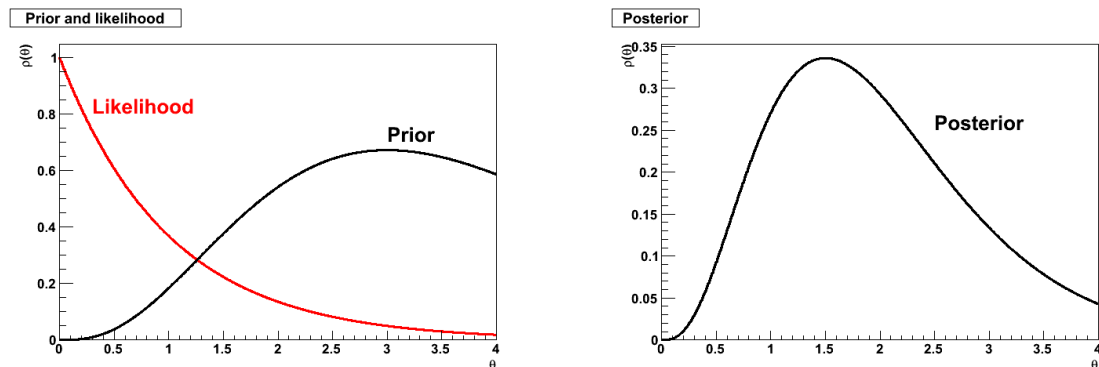
$$P(\text{Data}) = \sum_i P(\text{Data}, \theta_i) = \sum_i P(\text{Data}|\theta_i)P(\theta_i)$$

and for a continuous parameter

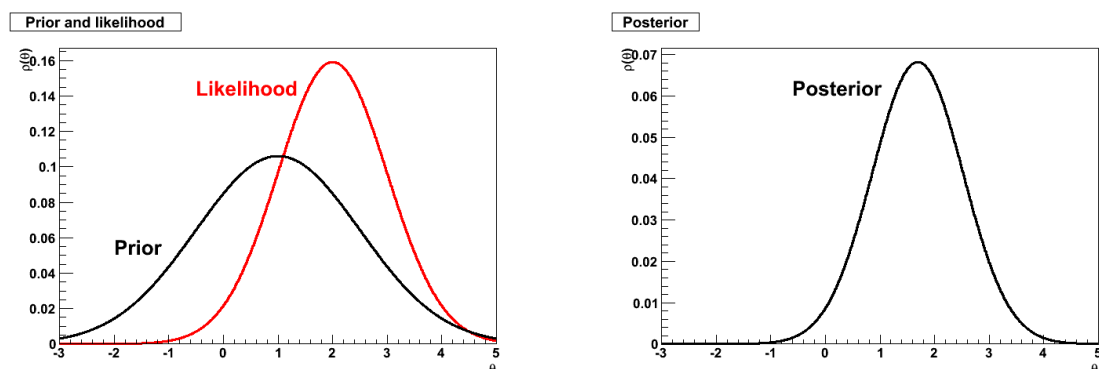
$$P(\text{Data}) = \int P(\text{Data}, \theta) d\theta = \int P(\text{Data}|\theta)P(\theta) d\theta$$

so it is simply the sum or integral of the numerator, which ensures that $P(\theta|\text{Data})$, the resulting probability distribution for θ , is normalised correctly. This property holds generally and is called the ‘marginalisation rule’ (also seen in Lecture 1).

You can think of the Bayesian equation as taking our existing knowledge of the parameter probability distribution $P(\theta)$, and updating it with an experiment to give a new probability distribution $P(\theta|\text{Data})$. The probability $P(\theta)$ therefore expresses our knowledge of the parameter before the experiment and hence is often called the ‘prior’, while the resulting probability $P(\theta|\text{Data})$ is called the ‘posterior’ and expresses our knowledge after the experiment. The likelihood can be thought of as updating our knowledge from the prior to give the posterior. A graphical example of the Bayesian equation is shown below.



A common case (due to the Central Limit Theorem) is that our prior knowledge (perhaps from a previous measurement) often has the form of a Gaussian distribution. The likelihood is often also Gaussian, as there is often a Gaussian distribution in the new data. Two Gaussian distributions multiplied together produce another Gaussian, so the posterior is also Gaussian distributed, as shown below.



The posterior is the probability distribution for the parameter after having seen the results of the experiment. If we did a second experiment later, then the posterior of the first experiment encapsulates the state of our knowledge about the parameter before the second experiment. Hence the first posterior becomes the second prior. This can be repeated many times.

Let's see an example. A friend of yours asks you to play a coin-tossing game where you bet on heads or tails, but he insists you use his coin and that he will bet heads. You suspect he will cheat and he has a double-headed coin. You play the game to determine if this is true or not (potentially at the cost of some money). The parameter to be determined is the number of heads on the coin, i.e. whether it is single-headed (SH) or double-headed (DH). You have to assign a prior probability for him cheating, i.e. for the coin being SH or DH. There is no ‘right’

numerical answer; this is a subjective Bayesian prior and so it is up to your judgement. Let's take the initial probability you estimate for him to be cheating to be 20%, which means

$$P(\text{DH}) = 0.2 \quad \text{so} \quad P(\text{SH}) = 0.8$$

You play twice and he wins with two heads in a row. The probability of the coin being double-headed (DH) is given by

$$P(\text{DH}|\text{Two heads}) = \frac{P(\text{Two heads}|\text{DH})P(\text{DH})}{P(\text{Two heads})}$$

For the right-hand side, then the likelihood is

$$P(\text{Two heads}|\text{DH}) = 1$$

as a double-headed coin will always give two heads in two tosses. For the denominator, using the marginalisation rule

$$\begin{aligned} P(\text{Two heads}) &= P(\text{Two heads}|\text{DH})P(\text{DH}) + P(\text{Two heads}|\text{SH})P(\text{SH}) \\ &= 1 \times 0.2 + 0.25 \times 0.8 = 0.4 \end{aligned}$$

where the probability of getting two heads when using a normal (single-headed) coin is $P(\text{Two heads}|\text{SH}) = (1/2) \times (1/2) = 1/4$. Applying Bayes' theorem then gives

$$P(\text{DH}|\text{Two heads}) = \frac{P(\text{Two heads}|\text{DH})P(\text{DH})}{P(\text{Two heads})} = \frac{1 \times 0.2}{0.4} = 0.5$$

and so your initial guess (the prior) of 0.2 is modified to a posterior of 0.5 given the observed data, i.e. two heads. You get more suspicious that he is cheating.

You are not sure about your friend yet, so you decide to play for another two coin tosses, i.e. you repeat the experiment and (surprise) the result is another two heads. The prior for the second experiment is the posterior we just calculated. The likelihood is identical, so the evidence becomes

$$\begin{aligned} P(\text{Two heads}) &= P(\text{Two heads}|\text{DH})P(\text{DH}) + P(\text{Two heads}|\text{SH})P(\text{SH}) \\ &= 1 \times 0.5 + 0.25 \times 0.5 = 0.625 \end{aligned}$$

Applying Bayes' theorem then gives

$$P(\text{DH}|\text{Two heads}) = \frac{P(\text{Two heads}|\text{DH})P(\text{DH})}{P(\text{Two heads})} = \frac{1 \times 0.5}{0.625} = 0.8$$

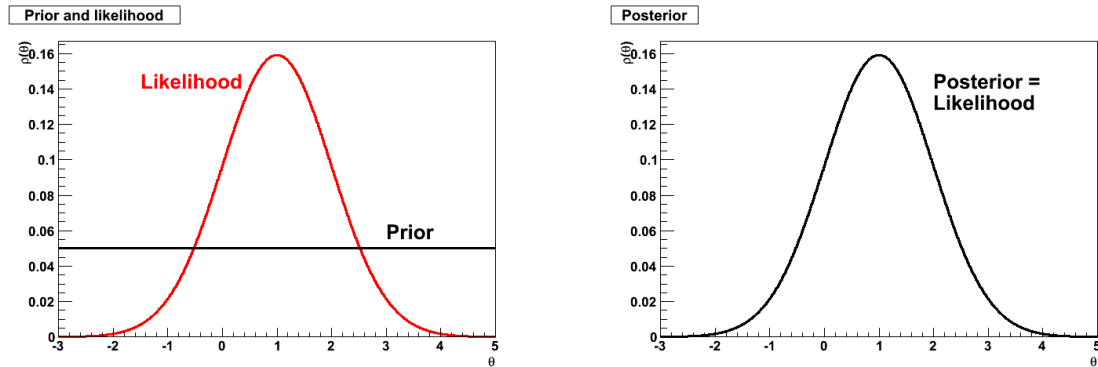
Now you definitely suspect him of cheating.

Note, if you calculated directly from your initial prior of $P(\text{DH}) = 0.2$, with the observed data of four heads (i.e. both experiments together), you would get exactly the same final answer.

30 Priors

It is with the prior that the problem with the Bayesian approach arises: we must have some knowledge of the parameter already. Sometimes this is OK; e.g. we may not be the first people to measure the electron mass. However, this is not always the case. In addition, it means we cannot give a result 'purely' from our own experiment to compare with other people's experiments.

If we can't, or don't want to, use previous results, then it is standard to use a 'flat' prior, i.e. a uniform distribution for $P(\theta)$. Here we just specify the upper and lower bounds, which



could be chosen conservatively to encompass a very large range. If we do choose a flat prior, then the posterior is the same shape as the likelihood.

Note that a flat prior might not always correspond to an unknown prior. In Lecture 3, we saw that a change in variables can change the shape of a distribution, so if we happened to use the square of the parameter instead, we would have to choose a different distribution in the parameter to get a flat distribution for the square of the parameter. Hence, the form of an unknown prior depends on the situation.

31 Relation to maximum likelihood

We discussed that the principle of maximum likelihood was a good way to get an estimate for a parameter. What does this estimation correspond to in terms of the Bayesian approach? As stated above, the likelihood appears directly in the expression we are using. Specifically

$$P(\theta|\text{Data}) \propto L(\theta)P(\theta)$$

where the likelihood has been evaluated for the particular experiment, as we did before. In particular, if we choose a flat prior so $P(\theta)$ is a constant, then

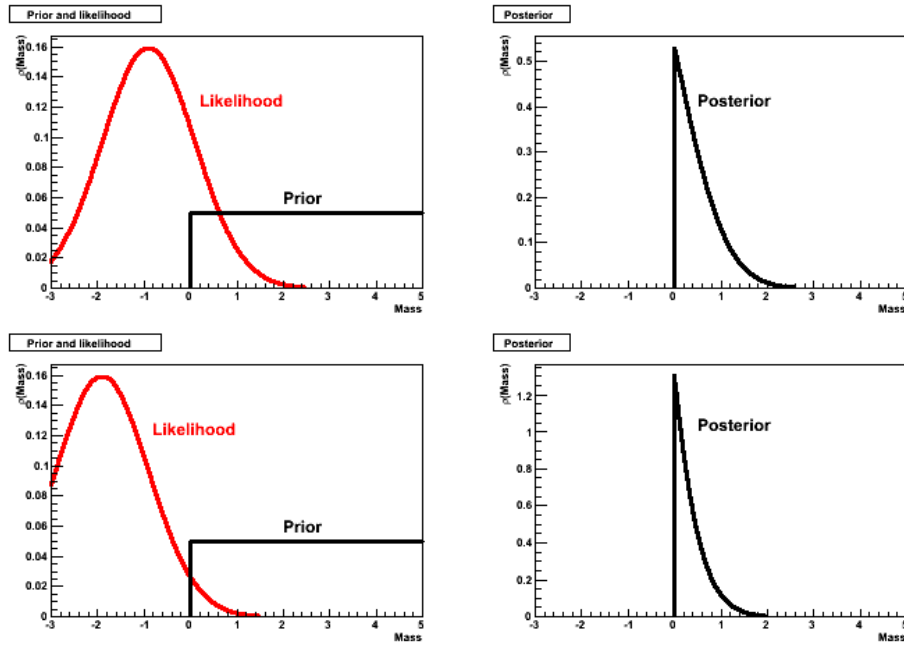
$$P(\theta|\text{Data}) \propto L(\theta)$$

Hence, the estimate obtained from the maximum likelihood approach corresponds to the most probable value of the parameter in the Bayesian approach when using a flat prior for the parameter. This in some sense ‘justifies’ the maximum likelihood principle, by making explicit which assumptions underpin it.

32 Physical constraints

We saw that frequentist methods have problems if they end up with results outside the physical region. We discussed an example of measuring the mass of a powder in the last lecture.

How does the Bayesian approach tackle this issue? The prior, whether flat or not, is the key; the probability of the mass genuinely being negative is of course zero, so that we simply make the prior be zero anywhere that is unphysical. This might leave only a small amount of the probability distribution in the physical region, but this is countered by the normalisation in the denominator (i.e. the evidence), which will also be small. Hence, it gives a sensible result overall, with the probability distribution for the parameter only being non-zero in the physical region. This is illustrated for the 99.1 g and 98.1 g mass measurements below (with a 100g weighing dish).



33 Frequentist or Bayesian?

This is a question which has vexed many people for a long time. Discussions can sometimes approach the intensity of religious wars; they are similar in that there is never any possibility of ‘proving’ one is better than the other.

Much of the time it doesn’t matter. A Gaussian distribution well away from any physical constraints gives the same result in both frameworks. Since the Central Limit Theorem says many distributions look Gaussian, then in most cases the difference is minimal. Even when there is a difference, it is perfectly correct to use either method, as long as you state clearly, and give justification for, what you did.

34 Acknowledgment

Many thanks to Mark Richards and other previous lecturers for their development of the Statistics of Measurement course.